NOVA INFORMATION MANAGEMENT SCHOOL

Business Intelligence II

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1. Business Context

The Hollywood film industry operates at the intersection of creativity and commercial strategy. Films are artistic products, but they are also high-risk investments that require careful financial planning. Studios and production companies invest heavily in development, casting, production, marketing, and distribution, often under conditions of significant uncertainty regarding audience reception and financial return. As a result, understanding the factors that drive commercial success has become increasingly critical in an industry characterized by volatility, intense competition, and rapidly changing consumer preferences.

MAD Investment Group is a Lisbon-based fictional company pioneering a new era of film investment. Formed by four investors with complementary backgrounds in data analysis and creative strategy, the team combines analytical intelligence with cultural sensitivity to guide cinematic decision-making. Lisbon, as a growing hub for creative industries and tech innovation, offers strategic access to European and international markets, making it an ideal base for a data-driven film investment firm.

The global film market is complex and highly competitive, with audience preferences shaped by factors such as casting, genre, release timing, and critical reception. Traditional decision-making often relies on experience and intuition, increasing uncertainty and financial risk. To address this challenge, MAD Investment Group applies Business Intelligence (BI) principles to film production by analyzing historical box-office data and audience behavior. This approach enables the identification of measurable success patterns and supports investment decisions grounded in evidence rather than intuition, blending creativity with analytical intelligence.

This system allows us to explore what truly drives cinematic success, blending creativity with intelligence. In short, it’s our way of bringing data and art together: creativity guided by intelligence, cinema powered by data.

“Truly creative things happen when one thinks differently, yet nobody wants to think differently.”

— Steven Soderbergh

2. Data Warehouse

The Data Warehouse developed by MAD Investment Group was designed to support evidence-based decision-making in film investment by enabling structured, high-performance analysis of box-office performance across time, content, talent, geography, and audience reception. The design of the warehouse was driven directly by the business needs defined in BI I and implemented using a dimensional modelling approach suitable for analytical workloads.

A dimensional modelling methodology based on a star schema was adopted. The core business process modelled is the daily box-office performance of films. The Data Warehouse is structured around a single fact table, fact_daily_box_office, supported by five conformed dimensions: dim_date, dim_film, dim_actor, dim_director, and dim_production_company.

The fact table captures two native measures: Daily Gross Revenue and Days Since Release. These measures directly support the evaluation of commercial success, growth dynamics, and long-term performance patterns. The grain of the fact table is **one record per Film × Date**.

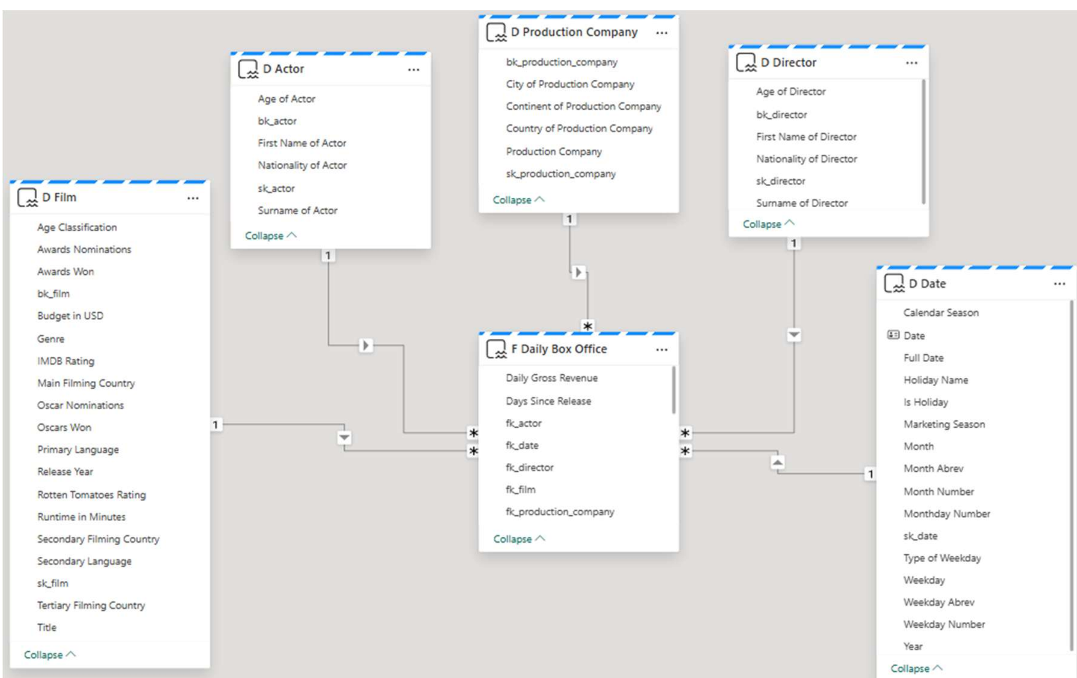

Figure 1: DW_MAD_MOVIES Star Schema

3. Revised Business Questions

This section presents the revised and finalized set of Business Questions that drive the reporting and dashboarding solutions developed in BI II. The questions have been reviewed, improved, and expanded relative to the original BI I set, incorporating feedback from the first delivery evaluation.

The 11 Business Questions are organized across two analytical levels: Descriptive (D1–D9) and Predictive (P1–P2).

3.1 Descriptive Analytics

Descriptive questions answer what happened and are fully addressed through DAX measures and Power BI visual reports.

> ***BQ D1 — What is the total and average daily box-office revenue by genre, release season, and age classification?***

This question aggregates gross revenue per day across the film, date, and classification dimensions. It enables MAD Investment Group to identify which genre–season–audience combinations consistently generate the highest returns, directly supporting strategic investment targeting.

> ***BQ D2 — Which lead actors and lead directors rank highest by cumulative box-office revenue across all their films?***

This question ranks talent by the total revenue generated across every film they appear in as lead actor or director. It provides MAD Investment Group with a data-driven foundation for casting and director selection decisions.

> **BQ D3 — How do average daily box-office revenue and audience ratings compare across runtime segments — under 90 minutes, 90 to 120 minutes, and over 120 minutes?**

This question compares financial performance and critical reception across three runtime tiers using average revenue and average IMDb and Rotten Tomatoes scores per segment. It informs production decisions around film length.

BQ D4 — What is the average weekly box-office revenue change by genre over the theatrical run?

This question measures the average week-on-week revenue variation per genre using a DAX calculated measure. It reveals which genres sustain audience interest over time and which decay quickly, guiding release and marketing timing decisions for MAD Investment Group.

BQ D5 — Which films rank among the top 5 highest-rated in each budget tier — high-budget versus low-budget — and how do their revenues compare?

This question segments the film catalogue by production budget and surfaces the top-rated titles in each group alongside their total box-office revenue. It allows MAD Investment Group to evaluate whether high artistic quality requires high production investment.

BQ D6 — Which weekdays generate the highest average daily box-office revenue, and how does revenue evolve across the days since release?

This question analyses audience behaviour patterns by day of week and by position in the theatrical lifecycle. It supports decisions on release day selection and the duration of marketing campaigns.

BQ D7 — How is total box-office revenue distributed across filming countries and production company continents?

This question maps cumulative revenue by primary filming location and by the geographical origin of the production company. It identifies regional production hubs and markets that consistently outperform, informing co-production and location strategy for MAD Investment Group.

BQ D8 — What is the average return on investment by film genre and production budget tier?

This question aggregates total box-office revenue divided by production budget across genre and budget tier groupings. It enables MAD Investment Group to identify which genre and budget combinations consistently deliver the highest financial return per dollar invested, directly supporting capital allocation decisions.

BQ D9 — Which lead actors generate the highest average box-office revenue per film across different genres?

This question ranks lead actors by their average box-office revenue per film, segmented by genre. Unlike D2, which measures cumulative revenue, this question normalizes performance by the number of films to identify actors who consistently deliver high returns regardless of volume. It directly supports casting decisions for MAD Investment Group by surfacing talent that maximizes average revenue per production.

3.2 Predictive Analytics

Predictive questions answer what is likely to happen. Both P1 and P2 are addressed using machine learning models detailed in section 5, where methodology, results, and integration with the Power BI report are fully described.

BQ P1 — Based on historical genre revenue trends, which genres are projected to grow or decline in the next release window?

This question uses a linear regression model built in a Microsoft Fabric PySpark notebook to project near-term revenue trajectories by genre. For each genre, the model fits a regression line over historical annual revenue and classifies the trend as Growing, Stable, or Declining based on the slope relative to the genre mean. Results are written to the ml_genre_trends Delta table in the Lakehouse and consumed directly in the Power BI report.

BQ P2 — Given a film's genre, runtime segment, release season, and age classification, what is its predicted box-office revenue tier — low, mid, or high?

Each film profile is classified into a predicted revenue tier (Low, Mid, or High) using a Random Forest classification model integrated via a Microsoft Fabric PySpark notebook. The model is trained on historical films with known box-office revenue and returns a predicted tier label alongside a confidence score. Both outputs are written to the ml_tier_predictions Delta table in the Lakehouse and consumed directly in the Power BI report, allowing MAD Investment Group to assess the revenue potential of a planned production before committing investment.

4. Semantic Model

This section documents the design, implementation, and configuration of the Semantic Model developed in Microsoft Fabric for the MAD Investment Group BI II project. The Semantic Model sits on top of the Data Warehouse built in BI I and serves as the analytical layer that connects raw data to the Power BI reporting and dashboarding solutions. It encompasses the table relationships inherited from the star schema, dimension hierarchies, eight DAX calculated measures, and four Key Performance Indicators that collectively answer the eleven Business Questions defined in section 3.

4.1 Model Overview and Table Relationships

The Semantic Model was built on top of the existing Data Warehouse (DW_MAD_MOVIES) without structural changes to the underlying tables. All relationships were carried over from the BI I star schema: each dimension table connects to the fact table F Daily Box Office through its respective surrogate key, establishing one-to-many relationships from dimension to fact. The model contains the following tables: F Daily Box Office (fact), D Film, D Date, D Actor, D Director, and D Production Company (dimensions).

All table and column names were updated in the Semantic Model to use readable, business-friendly labels. Technical identifiers such as surrogate keys and business keys were hidden from report view to ensure that end users interact only with analytically meaningful attributes.

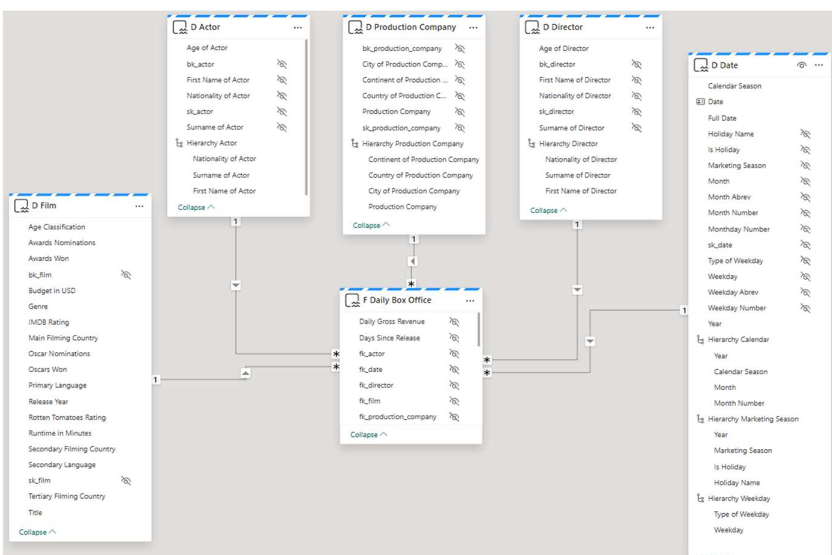

Figure 2: Star Schema with Hidden Technical Identifiers (Surrogate and Business Keys) and Hierarchies

4.2 Hierarchies

Six hierarchies were created across the dimension tables to support drill-down analysis in Power BI reports, distributed across four dimensions: D Date (three hierarchies), D Actor, D Director, and D Production Company. Each hierarchy reflects a meaningful analytical path that directly supports one or more of the Business Questions defined in section 3.

All six hierarchies are visible in the Semantic Model image in section 4.1 above (figure 2). Note that all individual attributes included as hierarchy levels were hidden from the default report view to avoid redundancy; users interact with the hierarchy as a whole rather than with its constituent columns individually.

4.2.1 Calendar Hierarchy — D Date

The Calendar Hierarchy provides standard temporal drill-down from year to individual date, supporting all time-based analyses including seasonal revenue comparisons, monthly trends, and day-level lifecycle analysis.

- Level 1: Calendar Year
- Level 2: Calendar Season (Winter, Spring, Summer, Fall)
- Level 3: Month
- Level 4: Date

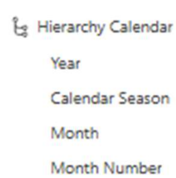

Figure 3: Calendar Hierarchy in the Semantic Model

4.2.2 Marketing Season Hierarchy — D Date

The Marketing Season Hierarchy captures the film industry's commercial release calendar, grouping dates by marketing window rather than astronomical season. This hierarchy directly supports BQ D1 and BQ D4 by enabling revenue analysis within industry-relevant release periods and revealing which marketing windows sustain or accelerate audience interest over the theatrical run.

- Level 1: Calendar Year

- Level 2: Marketing Season (Summer Break, Christmas Break, Halloween, Easter Break, Thanksgiving Break, Regular)
- Level 3: Holiday Name (Christmas Day, Independence Day, Halloween, etc.)

The is_holiday column was deliberately excluded from this hierarchy as it is redundant; the presence of a holiday name already implies the date is a holiday. It is retained in the model as a standalone filter column for slicer use.

Figure 4: Marketing Season Hierarchy in the Semantic Model

4.2.3 Weekday Hierarchy — D Date

The Weekday Hierarchy supports audience behaviour analysis by grouping dates from weekday type down to individual date. It directly answers BQ D6 by enabling users to compare revenue between weekdays and weekends, and then drill down to specific days of the week to identify patterns such as Friday release boosts or Sunday drops.

- Level 1: Type of Weekday (Weekday / Weekend)
- Level 2: Weekday (Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday)
- Level 3: Date

Figure 5: Weekday Hierarchy in the Semantic Model

4.2.4 Actor Hierarchy — D Actor

The Actor Hierarchy enables geographical and identity-based drill-down for talent analysis, supporting BQ D2 and BQ D9 by allowing users to navigate from nationality to individual actor.

- Level 1: Nationality of Actor
- Level 2: Surname of Actor
- Level 3: First Name of Actor

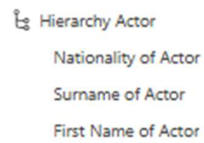

Figure 6: Actor Hierarchy in the Semantic Model

4.2.5 Director Hierarchy — D Director

The Director Hierarchy mirrors the Actor Hierarchy structure, supporting BQ D2 for director-level analysis.

- Level 1: Nationality of Director
- Level 2: Surname of Director
- Level 3: First Name of Director

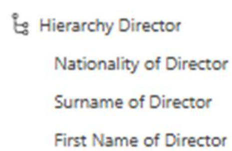

Figure 7: Director Hierarchy in the Semantic Model

4.2.6 Production Company Hierarchy — D Production Company

The Production Company Hierarchy provides a four-level geographical drill-down from continent to individual company, supporting BQ D7 and BQ D8 by enabling geographic production analysis and investment efficiency comparisons across production origins.

- Level 1: Continent of Production Company
- Level 2: Country of Production Company
- Level 3: City of Production Company
- Level 4: Production Company (name)

Figure 8: Production Company Hierarchy in the Semantic Model

4.3 Data Quality Corrections

During the Semantic Model development, several data quality issues carried over from the BI I enrichment phase were identified and corrected directly in the Data Warehouse. These corrections were necessary to ensure accurate grouping, consistent display in geographical visuals, and correct functioning of the dimension hierarchies.

First, nationality values for lead actors and directors contained inconsistent spellings, abbreviations, and encoding errors introduced during the API enrichment phase. Multiple variants referring to the same country were standardized to a single consistent form to ensure that the Actor and Director Hierarchies group correctly and that geographical analyses produce accurate results.

Examples of corrections applied: “USA]”, “U.S.”, “US”, and “United States” were all consolidated to “USA”; “india” was corrected to “India”; and “California” was re-mapped to “USA” as it referred to a state rather than a country.

Second, the production company country values were stored as two-letter ISO codes rather than full country names. These were expanded to their complete written form to ensure consistent display in geographical visuals and to support correct drill-down behavior in the Production Company Hierarchy. For example, “JP” was expanded to “Japan”, “GB” to “England”, and “US” to “USA”.

4.4 DAX Calculated Measures

All DAX measures were created inside the F Daily Box Office fact table in the Semantic Model. This is consistent with dimensional modelling best practice: measures are anchored to the grain of the fact table so that filter context flows correctly from all dimensions. The measures are organized into four groups: base measures, supporting measures, advanced measures, and KPI measures.

4.4.1 Base Measures

The following eight base measures form the foundation of the analytical layer. Every supporting, advanced, and KPI measure depends on one or more of these.

Total Revenue

Computes the sum of all daily gross revenue in the current filter context. This is the primary financial measure used across all revenue-based Business Questions (D1, D2, D5, D7, D8, D9).

```
Total Revenue = SUM( 'F Daily Box Office'[Daily Gross Revenue] )
```

Figure 9: Total Revenue DAX Measure

Average Daily Revenue

Computes the average revenue per day across the filtered context. Used for runtime segment comparisons (D3) and weekday pattern analysis (D6).

```
Average Daily Revenue = AVERAGE( 'F Daily Box Office'[Daily Gross Revenue] )
```

Figure 10: Average Daily Revenue DAX Measure

Total Films

Returns the count of distinct films in the current filter context using DISTINCTCOUNT on the foreign key. Used for portfolio analysis and genre-level comparisons (D1, D5, D9).

```
Total Films = DISTINCTCOUNT( 'F Daily Box Office'[fk_film] )
```

Figure 11: Total Films DAX Measure

Average IMDb Rating

Computes the average IMDb score across films in context. The AVERAGEX and VALUES pattern is used to ensure each film is counted once regardless of how many daily fact rows it has, avoiding double-counting caused by the repeated rows in the fact table. Supports runtime (D3) and budget tier (D5) analyses.

```
Average IMDb Rating =
AVERAGEX(
    VALUES( 'D Film'[Title] ),
    MAX( 'D Film'[IMDB Rating] )
)
```

Figure 12: Average IMDb Rating DAX Measure

Average RT Rating

Computes the average Rotten Tomatoes score using the same AVERAGEX and VALUES pattern as the IMDb measure. Supports D3 and D5.

```
Average RT Rating =
AVERAGEX(
    VALUES( 'D Film'[Title] ),
    MAX( 'D Film'[Rotten Tomatoes Rating] )
)
```

Figure 13: Average RT Rating DAX Measure

Average Days Since Release

Computes the average number of days since a film's premiere across the filtered context. Used in theatrical lifecycle analysis (D4, D6).

```
Average Days Since Release = AVERAGE( 'F Daily Box Office'[Days Since Release] )
```

Figure 14: Average Days Since Release DAX Measure

Revenue per Budget Dollar

Computes the return on investment ratio by dividing total box-office revenue by the total production budget across all films in the current filter context. The SUMX and VALUES pattern avoids double-counting budget values across the repeated rows of the fact table. DIVIDE handles division by zero gracefully. Foundation of BQ D8.

```
Revenue per Budget Dollar =
DIVIDE(
    [Total Revenue],
    SUMX(VALUES('D Film'[Title]), MAX('D Film'[Budget in USD]))
)
```

Figure 15: Revenue per Budget Dollar DAX Measure

Average Revenue per Film

Computes the average box-office revenue per film by dividing Total Revenue by the count of distinct films. Unlike Total Revenue, this measure normalizes performance by production volume, providing a fairer comparison across actors and countries. DIVIDE handles division by zero gracefully. Foundation of BQ D9.

```
Avg Revenue per Film =
DIVIDE(
    [Total Revenue],
    [Total Films]
)
```

Figure 16: Average Revenue per Film DAX Measure

Total Revenue (USD)

Computes total box-office revenue with explicit US-dollar formatting, used where the currency must be shown unambiguously, chiefly the paginated Talent league tables reports (D2), whose print-ready league tables display this column as "Total Revenue (USD)".

```
Total Revenue USD = SUM( 'F Daily Box Office'[Daily Gross Revenue] )
```

Figure 17: Total Revenue (USD) DAX Measure

4.4.2 Supporting Measures

The following three measures serve exclusively as reference values for the KPI calculations. They are hidden from the report view and are not used directly in visuals.

Average Genre Revenue

Computes the average total revenue across all genres in the current filter context using AVERAGEX over genre values. Reference value for KPI 1.

```
Avg Genre Revenue =
AVERAGEX ( ALL ( 'D Film'[Genre] ), [Total Revenue] )
```

Figure 18: Average Genre Revenue Supporting DAX Measure

Average Actor Revenue

Computes the average total revenue across all lead actors in the current filter context using AVERAGEX over actor values. Reference value for KPI 3.

```
Avg Actor Revenue =
CALCULATE(
    AVERAGEX(VALUES('D Actor'[bk_actor]), [Total Revenue]),
    REMOVEFILTERS('D Actor')
)
```

Figure 19: Average Actor Revenue Supporting DAX Measure

Average Genre Days Since Release

Computes the average days since release across all genres in the current filter context using AVERAGEX over genre values. Reference value for KPI 4.

```
Avg Genre Days Since Release =
AVERAGEX(VALUES('D Film'[Genre]), [Average Days Since Release])
```

Figure 20: Average Genre Days Since Release Supporting DAX Measure

4.4.3 Advanced Measures

The following three measures support the more complex theatrical lifecycle analysis required by BQ D4.

Revenue Last Week

Computes total box-office revenue for the previous week using DATEADD. Supporting measure for Weekly Revenue Change.

```
Revenue Last Week =
CALCULATE(
    [Total Revenue],
    DATEADD( 'D Date'[Date], -7, DAY )
)
```

Figure 21: Revenue Last Week Advanced DAX Measure

Weekly Revenue Change

Computes the week-on-week revenue difference by subtracting Revenue Last Week from Total Revenue. Used in theatrical lifecycle analysis (D4).

```
Weekly Revenue Change = [Total Revenue] - [Revenue Last Week]
```

Figure 22: Weekly Revenue Change Advanced DAX Measure

Average Weekly Revenue Change

Computes the average week-on-week revenue variation across genres using AVERAGEX. Directly answers BQ D4 by quantifying revenue decay or growth momentum by genre over the theatrical run.

```
Avg Weekly Revenue Change =
AVERAGEX(
    VALUES( 'D Film'[Genre] ),
    [Weekly Revenue Change]
)
```

Figure 23: Average Weekly Revenue Change Advanced DAX Measure

4.4.4 KPI Measures

Four Key Performance Indicators were developed as required by the project handout. All KPIs compare current performance against a reference value and are designed to be displayed using Power BI's native KPI visual.

KPI 1 — Revenue vs Genre Average

Measures how a selected genre's total revenue compares against the average revenue across all genres, expressed as a percentage deviation. A positive value indicates above-average performance; a negative value indicates below average. Directly supports BQ D1 by signaling whether a genre is a commercially promising investment target for MAD Investment Group.

```
KPI Days vs Genre Avg =
VAR AvgG = CALCULATE([Avg Genre Days Since Release], REMOVEFILTERS('D Film'[Genre]))
RETURN
IF(
    HASONEVALUE('D Film'[Genre]),
    DIVIDE([Average Days Since Release] - AvgG, AvgG),
    BLANK()
)
```

Figure 24: Revenue vs Genre Average KPI DAX Measure

KPI 2 — Return on Investment (ROI)

Measures the financial return generated per dollar of production budget invested by dividing total revenue by total production budget. The SUMX and VALUES pattern avoids double-counting budget values across fact table rows. Directly supports BQ D8 by enabling comparison of investment efficiency across genres and budget tiers.

```
KPI ROI =
DIVIDE(
    [Total Revenue],
    SUMX(VALUES('D Film'[sk_film]), CALCULATE(MAX('D Film'[Budget in USD])))
)
```

Figure 25: Return on Investment KPI DAX Measure

KPI 3 — Revenue vs Average Actor Revenue

Measures how a selected actor's total revenue compares against the average revenue across all lead actors, expressed as a percentage deviation. A positive value indicates above-average performance; a negative value indicates below average. Directly supports BQ D2 and BQ D9 by signaling whether a lead actor consistently outperforms the market, supporting casting decisions for MAD Investment Group.

```
KPI Revenue vs Actor Avg =
IF(
    HASONEVALUE('D Actor'[bk_actor]),
    DIVIDE([Total Revenue] - [Avg Actor Revenue], [Avg Actor Revenue]),
    BLANK()
)
```

Figure 26: Revenue vs Actor Average KPI DAX Measure

KPI 4 — Days Since Release vs Genre Average

Measures how long a film has been in theatrical release compared to the average run length of films in the same genre, expressed as a percentage deviation. A positive value indicates longer-than-average retention; a negative value indicates faster decay. Directly supports BQ D6 by assessing audience retention relative to genre expectations, informing marketing campaign and release strategy decisions.

```
KPI Days vs Genre Avg =
VAR AvgG = CALCULATE([Avg Genre Days Since Release], REMOVEFILTERS('D Film'[Genre]))
RETURN
IF(
    HASONEVALUE('D Film'[Genre]),
    DIVIDE([Average Days Since Release] - AvgG, AvgG),
    BLANK()
)
```

Figure 27: Days vs Genre Average KPI DAX Measure

5. Predictive Analytics — Methodology & Results

As part of the analytical layer of this project, we developed two predictive models in the NB_Predictive_BQ notebook (located in 2. ETL WORK → PREDICTIVE ANALYTICS folders, since the notebook produces data outputs consumed downstream in the same way as ETL loads) to answer forward-looking business questions about box-office performance. Both models were built in Python using the scikit-learn library, running in a Microsoft Fabric PySpark notebook that reads directly from the DW_MAD_MOVIES SQL data warehouse. The notebook trains the models and writes the outputs as Delta tables to the Lakehouse (LH_SOURCES_MAD_MOVIES). These tables (ml_genre_trends and ml_tier_predictions) are automatically exposed via the Lakehouse SQL analytics endpoint and added as sources in the Semantic Model, allowing the predicted tier labels, confidence scores, genre trends and projected revenues to be consumed directly in the Power BI report without requiring structural changes to the data warehouse or the creation of new fact tables. The two predictive Business Questions (P1 and P2) are therefore answered using the results generated by this notebook rather than DAX measures.

5.1 BQ P1 — *Which film genres are projected to grow or decline in the next release window?*

Approach: The goal was to identify which film genres have been gaining or losing box office revenue over time, and to project where each genre is headed in the next release window.

We extracted total box-office revenue per genre per year from the data warehouse, joining fact_daily_box_office, dim_film and dim_date. This produced an annual revenue time series for each of the 193 genres in the dataset, spanning 2022–2024. For each genre a simple linear regression was fitted (year as input, annual revenue as output); the slope of the fitted line is the key output:

- If the slope is above +5% of the genre's mean annual revenue → “Growing”
- If the slope is below -5% → "Declining"
- Otherwise → "Stable"

The line was then extended one year beyond the last observed point to produce a projected revenue figure, and we also recorded n_years (the number of annual data points behind each genre's trend)

A data limitation we record and handle. The warehouse holds only three years of box-office history, so genres differ greatly in how much data supports their trend: of the 193 genres, only 56 have all three years, while the rest have one or two. A genre with a single year cannot have a slope and is set to Stable; a genre with two years rests on only two points. For reliability, we therefore filter the P1 report visuals to n_years ≥ 3. The effect is material: across all genres the split is dominated by single-year "Stable" genres, but on the reliable (n_years ≥ 3) basis it becomes 28 Growing, 25 Declining and 3 Stable, so the apparent "stable majority" is an artefact of thin history, not a market signal.

Results: The charts below show the linear-regression trend lines and projections for five representative growing genres (AnimationAdventureComedy, ActionAdventureComedy, ComedyDramaRomance, Comedy and Horror).

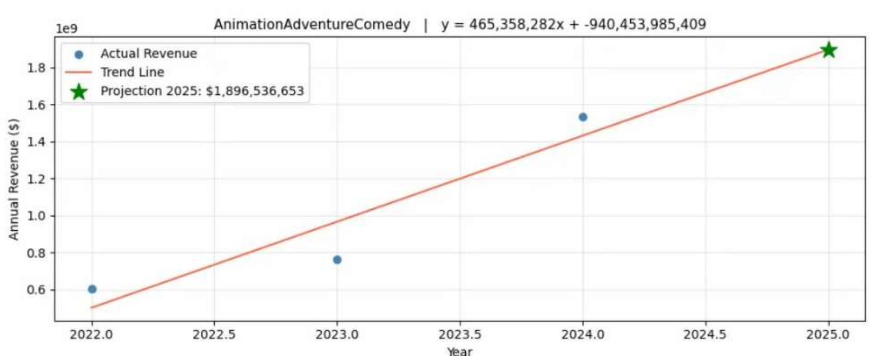

Figure 28: AnimationAdventureComedy — projected $1.9B

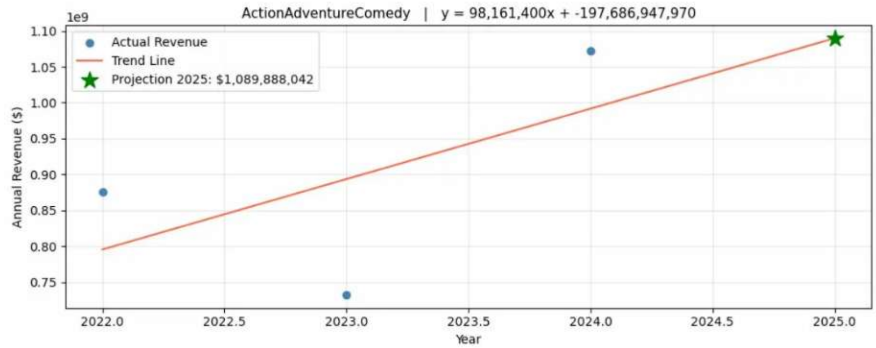

Figure 29: ActionAdventureComedy — projected $1.1B

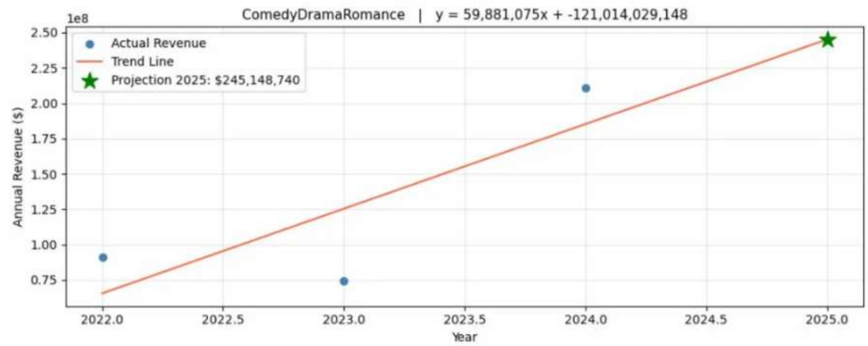

Figure 30: ComedyDramaRomance — projected $245M

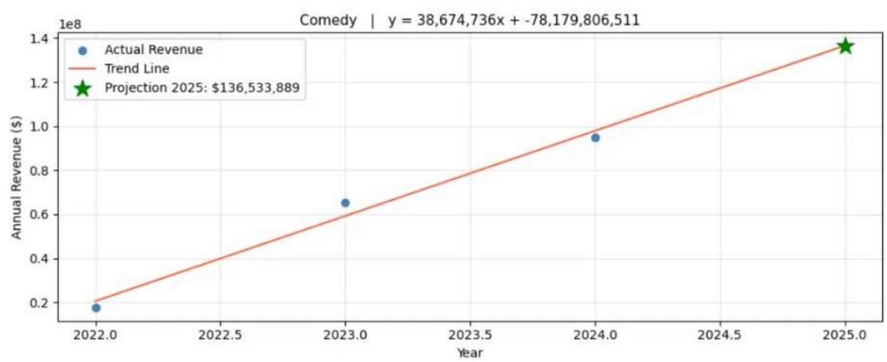

Figure 31: Comedy — projected $136M

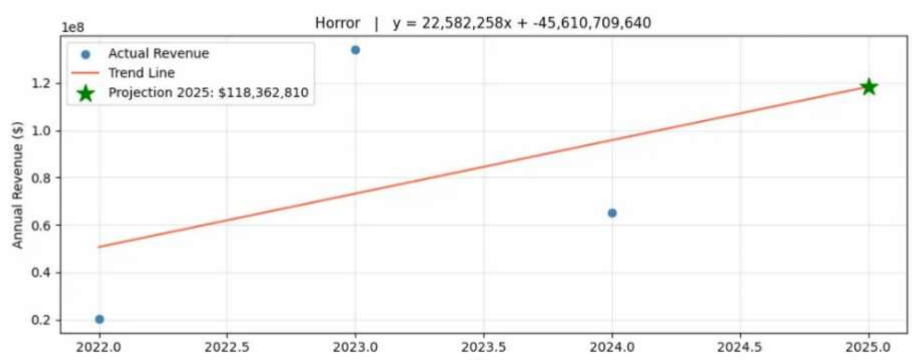

Figure 32: Horror — projected $118M

Each chart shows the actual annual revenue (blue dots), the fitted regression line (red), and the projection (green star); the regression equation in each title (y = slope × year + intercept) quantifies the yearly rate of change.

Results were written to the ml_genre_trends Delta table (193 genres, each with its trend, projected revenue, slope % and n_years) and are directly accessible from Power BI.

5.2 BQ P2 — *Given a film's characteristics, what is its predicted box office revenue tier?*

Approach: The goal was to build a model capable of predicting, before a film's revenue is known, whether it will be a High, Mid or Low box-office performer - directly useful for investment planning and release strategy.

<u>Step 1 — Defining the revenue tiers:</u>

Revenue tiers were defined using the 33rd and 66th percentiles of total gross revenue across all 782 historical films in the dataset:

- Below the 33rd percentile (under $340,199) → Low
- Between the 33rd and 66th percentile (up to $6,975,094) → Mid
- Above the 66th percentile → High

```
Films loaded:       782
Films with revenue: 782
Seasons found:      ['Fall', 'Spring', 'Summer', 'Winter']

Q33: 340,199  |  Q66: 6,975,094
```

Figure 33: Model Input Validation — Films Loaded and Revenue Tier Thresholds

By construction the true tier labels are near-balanced — 266 High, 258 Mid, 258 Low — which keeps the model from becoming biased toward any single class.

```
Tier distribution (historical):
tier
High    266
Mid     258
Low     258
Name: count, dtype: int64
```

Figure 34: Historical Tier Distribution — Training Dataset

Step 2 — Feature selection:

The following film characteristics were selected as model inputs:

- Film genre: type of film (e.g. Drama, Horror, Comedy)
- Age classification: audience rating (e.g. PG-13, R, PG)
- Release season: season of first box office entry (Spring, Summer, Fall, Winter), derived from dim_date using each film's earliest box office record
- Runtime: length of the film in minutes, kept as a continuous numeric feature to preserve variance for tree-based splits
- Production budget: budget in USD
- IMDB rating: audience score
- Rotten Tomatoes rating: critic score
- Awards won and nominations: measure of critical recognition

Categorical features (genre, age classification, release season) were one-hot encoded to avoid imposing false ordinal relationships, producing a 217-column feature matrix; missing numeric values were imputed with each column's median.

Step 3 — Model training and evaluation:

Two classification algorithms were evaluated (Random Forest and Gradient Boosting) using 5-fold stratified cross-validation. This technique splits the data into 5 equal parts, trains the model on 4 parts and tests it on the remaining 1, repeating this process 5 times to produce a reliable and unbiased estimate of real-world performance.

Results:

Model	Train Accuracy	Test Accuracy	F1 Score (weighted)
Random Forest	0.787	0.719	0.716
Gradient Boosting	0.991	0.738	**0.737**

Table 1: Cross-Validation Results — Random Forest vs Gradient Boosting

Gradient Boosting was selected as the winner on cross-validated (held-out) weighted F1 (0.737 vs 0.716) and accuracy (0.738 vs 0.719), comfortably above the 0.60 target. We note its larger train–test gap (0.991 vs 0.738) compared with Random Forest's (0.787 vs 0.719): Gradient Boosting fits the training data harder, but because both the selection metric and every reported figure are computed on held-out folds rather than on the training data, its higher held-out performance is what justifies the choice. Random Forest is the close runner-up.

Step 4 — Feature importance:

The model provides insight into which features drove its predictions most strongly:

Feature	Importance
Film budget	51.2%
Runtime	4.2%
Awards nominations	4.2%
Awards won	3.9%
IMDB rating	3.8%
Rotten Tomatoes rating	1.6%
Film release year	1.4%

Table 2: Random Forest Feature Importance — Top Predictors

Production budget is by far the strongest predictor (≈ 51% of total importance), which is intuitive: higher-budget films benefit from wider releases and larger marketing campaigns. Runtime, the awards measures (nominations and wins) and the IMDB rating follow as secondary signals, with the Rotten Tomatoes rating and release year contributing smaller amounts. Together these numeric features account for roughly 70% of total importance; the remaining ~30% is distributed across the one-hot-encoded genre, age-classification and release-season columns, which are individually small (the largest single one being the "Unknown" age-classification flag at ≈ 1.4%) but collectively meaningful.

Results: Predictions and confidence for the historical set are computed out-of-fold (each film is scored only by the cross-validation folds that excluded it) giving an average prediction confidence of 82.5% across all 782 films. This sits slightly above the model's test accuracy (≈ 74%), reflecting that the selected model produces fairly peaked class probabilities, and well above the 33.3% random baseline for a three-class problem. Confidence (the certainty the model attaches to each prediction) and accuracy (its validated hit rate) are reported separately.

```
-- Predictions (out-of-fold) --  Avg confidence: 0.825
        film_title               film_genre predicted_tier  tier_confidence
               '83        CrimeDramaHistory            Low           0.5671
              1992 AnimationAdventureComedy            Low           0.5210
              2073      DocumentaryThriller            Mid           0.8007
        2nd Chance              DramaFamily            Low           0.9761
                65    ActionAdventureSci-Fi            Low           0.6949
       80 for Brady        ComedyDramaSport           High           0.8396
          A Banquet    HorrorMysteryThriller            Low           0.9880
           A Chiara                    Drama            Low           0.9921
A Complete Unknown      BiographyDramaMusic           High           0.7724
   A Different Man      ComedyDramaThriller            Mid           0.8538
```

Figure 35: Sample Tier Predictions — Predicted Revenue Tier and Confidence Score per Film

Tier predictions and confidence scores were written to the ml_tier_predictions Delta table in the Lakehouse, with a final predicted-tier distribution of 312 High, 249 Mid, and 221 Low, covering all 782 films and making them directly accessible from Power BI for visualisation and reporting.

5.3 Output & Power BI Integration

Both predictive models were successfully executed and their outputs written as Delta tables to the Lakehouse (LH_SOURCES_MAD_MOVIES), as shown below:

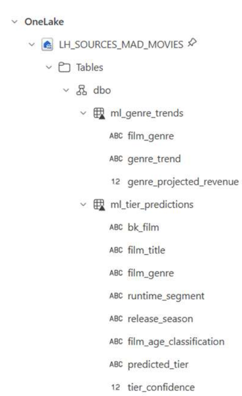

Figure 36: Delta Tables in the Lakehouse — ml_genre_trends and ml_tier_predictions

Table	Content	BQs
ml_genre_trends	Genre-level trend classification (Growing / Stable / Declining) and projected next-year revenue	BQ P1
ml_tier_predictions	Per-film predicted revenue tier (High / Mid / Low) and model confidence score	BQ P2

Table 3: Predictive Model Output Tables in the Lakehouse

Both tables are automatically exposed via the Lakehouse SQL analytics endpoint and are directly accessible from the Power BI report built within Microsoft Fabric, keeping the data warehouse schema untouched.

6. Reporting & Dashboarding

Data visualization is the final, value-delivering step of the business-intelligence chain. Everything built in the previous chapters (the data sources, the ETL, the data warehouse and its data marts) exists to be consumed, and it is only at the reporting layer that this engineering is turned into a decision. However sound the work upstream, its value is lost if the final reporting layer fails to communicate. This chapter therefore treats reporting not as a presentation afterthought but as the point at which the platform actually delivers value to MAD Investment Group, and every design choice that follows is judged by that standard: does it help the investment committee reach a decision?

6.1 Reporting strategy and design

This section explains how the eleven Business Questions were turned into reporting artefacts, the technical environment in which they were built, the interaction model offered to the end user, and the data-visualization principles that governed every design choice.

6.1.1 Reporting architecture: organizing by decision domain

We split the reporting layer into **three standard Power BI reports**, each scoped to a single decision domain rather than mixing themes, plus a set of **paginated reports**:

- **Report A - Performance & Return**: covers which film, budget, and talent characteristics drive revenue and return (D1, D3, D5, D8, D9).
- **Report B - Timing & Geography**: covers when and where revenue is generated (D4, D6, D7).
- **Report C - Predictive Analytics**: covers the forward-looking questions (P1, P2).
- **Paginated reports - Talent league tables**: rank the lead actors and lead directors by cumulative box-office revenue (D2, reinforcing D9).

Within each report we placed **one page per Business Question** (or per tightly related pair), so that every page is the visual answer to a single decision, and we positioned each KPI on the page of the domain it benchmarks. The pages are tied together with hub-and-spoke navigation: an Overview hub with headline cards and a master visual, and a return button on every content page.

The table below maps each Business Question to the artefact, page, and primary visuals that answer it, demonstrating full coverage of the eleven questions:

BQ	Report → Page	Primary visual(s)	KPI
D1	A → Genre, Season & Audience	Total/Avg daily revenue by genre, season, age classification (Top-N)	KPI 1: Revenue vs Genre Avg
D3	A → Runtime & Ratings	Combo: Avg daily revenue + IMDb + Rotten Tomatoes by Runtime Segment; detail table	—
D5	A → Top films by tier (sub-page)	Top-5 films by IMDb with Budget Tier slicer	—
D8	A → Budget & Return	ROI matrix (Top-10 genres × Budget Tier, heat-map) + ROI bar	KPI 2: ROI
D9	A → Talent Value	Avg revenue per film by lead actor (Top-15, ≥2 films); volume-vs-value combo; detail table	KPI 3: Revenue vs Actor Avg
D4	B → Weekly Momentum	Revenue by month (Top-5 genres) line + genres by avg weekly revenue change	—
D6	B → Release Timing & Audience	Revenue by weekday columns + decay curve (≤70 days)	KPI 4: Days vs Genre Avg
D7	B → Geography of Production	Filled map by filming country + bar by production-company continent + table	—
P1	C → Genre Trends	Projected genre revenue trend lines	—
P2	C → Predicted Tier	Predicted revenue-tier composition (Low/Mid/High) with confidence	—
D2	Paginated	Lead actors and lead directors ranked by cumulative revenue	—

Table 4: Each of the eleven Business Questions mapped to the report, page, primary visual(s), and KPI that answers it.

6.1.2 Paginated versus interactive Power BI reports

A paginated report (Power BI Report Builder) was used only for D2, where the deliverable is a long, ordered, pixel-perfect ranking meant to be read row-by-row and printed or exported for the investment committee: the lead-actor and lead-director league tables by cumulative revenue. Everything that is exploratory, filterable, and visual went into an interactive Power BI report. In short: ranked tabular listings destined for print/export are paginated; analytical, interactive exploration is delivered in Power BI.

6.1.3 Implementation environment

All reports were built with Power BI inside **Microsoft Fabric** and published to the *BI 2025 Group 181 Movies* workspace. The interactive reports run on the Direct Lake semantic model layered over the DW_MAD_MOVIES Warehouse and LH_SOURCES_MAD_MOVIES Lakehouse, so visuals query the Lakehouse data directly without an import refresh. Two properties of this environment shaped the entire design. First, Direct Lake forbids DAX calculated columns and tables and field parameters, so every binning and ordering attribute was created upstream in SQL and exposed through the model rather than computed in DAX. Second, the Fabric **web** report editor lacks *Edit interactions*, native drill-through configuration, and field parameters.

6.1.4 Interaction and navigation model

Each report is self-service and navigable. A Year slicer is present on the majority of the pages of the three reports and a Genre slicer only on genre-based pages; within a page, visuals cross-highlight on selection. Three interactions were engineered specifically to compensate for the missing web-editor features. A slicer is isolated to a single visual by placing both on a dedicated page reached by a navigation button, since a slicer affects only its own page. Native drill-through is replaced by a navigation button to a dedicated detail page that holds the underlying detail table, its own slicer, and a Back button. And contextual detail is delivered through custom tooltip pages shown on hover. Together these let the committee move from headline figures to film-level detail without leaving the report, while keeping each analytical page uncluttered.

6.1.5 Data-visualization principles

Page layout follows reading order: the headline KPI cards, when applied, and the primary visual occupy the top-left of each page, with supporting detail below. Related visuals are grouped by **proximity** and set apart with card enclosures, and color similarity is reused across pages, so the same hue always carries the same meaning, applying the Gestalt grouping principles. On top of this layout discipline, every visual was designed against the five main aspects of a visualization:

1. **Chart type, by purpose.** Comparison → bar/column; trend → line; composition → stacked/composition visuals; spatial → filled map; relationship and volume-versus-value → combo; deviation → KPI cards; ranking → Top-N bars and paginated tables.

2. **Color scheme.** A diverging/alert semantic applied consistently (green = above benchmark, red = below), sequential color saturation on the filled map, and a limited categorical palette for the Top-5 genre legends.

3. **Scale**. Consistent currency formatting; the decay curve capped at ≤ 70 days so the time axis stays meaningful; Top-N used to bound the genre axis, which holds roughly 193

concatenated values; and, on the genre-trend page, projections restricted to genres with at least three years of data, since genres with fewer years force the regression toward a flat "stable" reading and would otherwise manufacture a misleading stable majority.

4. **Description and context.** Titles written as the takeaway rather than the chart label, KPI cards that benchmark each value against an average, and custom tooltips that add context on hover.

5. **Clutter.** Hub-and-spoke navigation to keep pages clean, minimal slicers (Year everywhere, Genre only where the analysis is genre-based), no chart junk, and Top-N filtering instead of plotting every category.

6.2 Standard Report A: Performance & Return

Report A answers the questions about which film, budget, and talent characteristics drive revenue and return (D1, D3, D5, D8, D9) and hosts KPIs 1, 2, and 3. It is organized as a five-page hub-and-spoke report, and the design intent on every page is the same: choose the visual that makes the decision fastest to read, and keep the page clean enough that the answer is obvious.

Page 1: Overview. The hub page presents four headline cards (Total Revenue, Average ROI, Average Revenue per Film, and Total Films) above a master revenue visual, with navigation buttons to each themed page. Cards are the right visual here because the committee opens the report to check a few headline figures before anything else, and the hub-and-spoke structure then lets them drill into one domain at a time rather than meeting every visual at once. The ROI card is shown without conditional color, as a neutral headline figure, because at portfolio level there is no benchmark to beat; the color-coded ROI lives on the Budget page, where a comparison is meaningful. The page's purpose is to give the scale of the portfolio at a glance before drilling into any single domain.

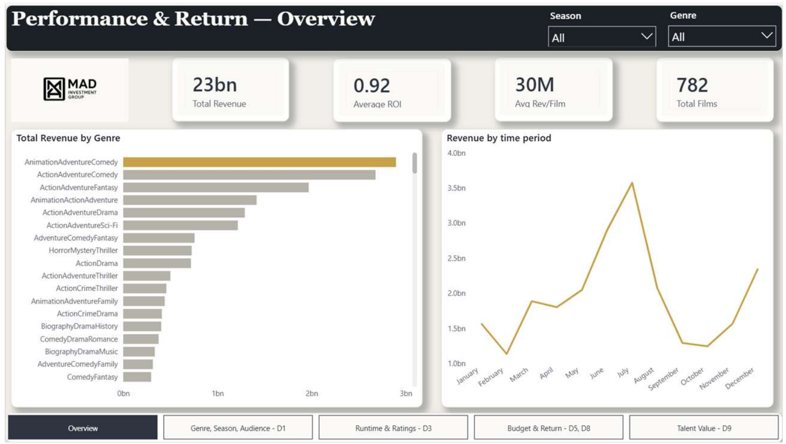

Figure 37: Report A (Performance & Return) - Overview page.

Page 2: Genre, Season & Audience (D1). This page answers D1 with visuals of total and average daily revenue broken down by genre, by calendar season, and by age classification. Because the genre attribute is a concatenated string with roughly 193 distinct values, the genre visual is always bounded with Top-N and a slicer rather than plotting every value: plotting all 193 would be unreadable, whereas Top-N keeps the chart legible while the slicer still allows any genre to be inspected. A bookmark toggle lets the committee switch these visuals between Total Revenue and Average Daily Revenue in place (the first showing which combinations earn most in aggregate, the second which earn most per screening day) so two distinct readings of D1 are delivered on one uncluttered page instead of duplicating every visual. The page carries KPI 1 - Revenue vs Genre Average as a conditional-format card (green at or above the genre average, red below), read for the genre currently selected; since the comparison is defined only for a single genre, the card stays blank until a genre is chosen in the slicer, which is the honest behavior rather than a fault.

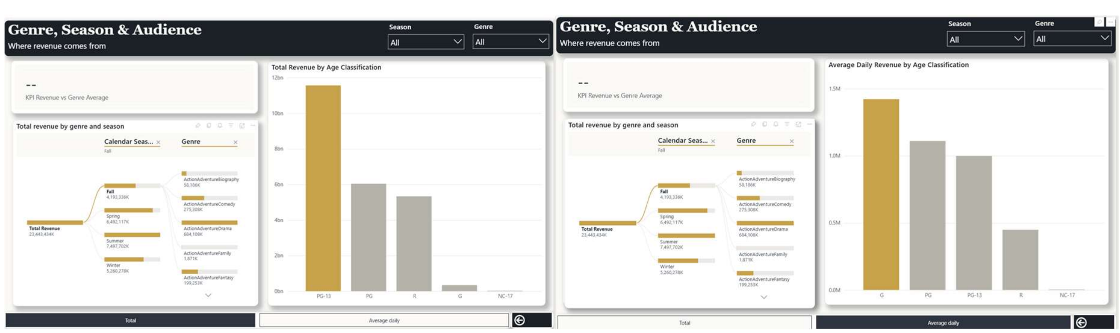

Figure 38: Report A (Performance & Return) - Genre, Season & Audience (D1) page.

Page 3: Runtime & Ratings (D3). D3 is answered with a combo chart that plots average daily revenue as columns against IMDb and Rotten Tomatoes scores as lines, all by Runtime

Segment (under 90, 90–120, over 120 minutes), supported by a detail table. A combo chart is the right choice because D3 is a relationship question across two different measures on the same axis (revenue against ratings by runtime) and the dual-axis form lets both be read together at a glance without separate charts. Two interactions enrich the page while keeping it uncluttered: a custom tooltip page (D3 Tooltip) shown on hover for context, and a button-driven drill-through substitute: a "Film details →" button navigates to a hidden segment-detail page holding a Title/Runtime/IMDb/RT/Revenue table with a Runtime Segment slicer and a Back button. This replaced native drill-through, whose field well rejected the configuration in the web editor, so film-level detail is available on demand without crowding the main page.

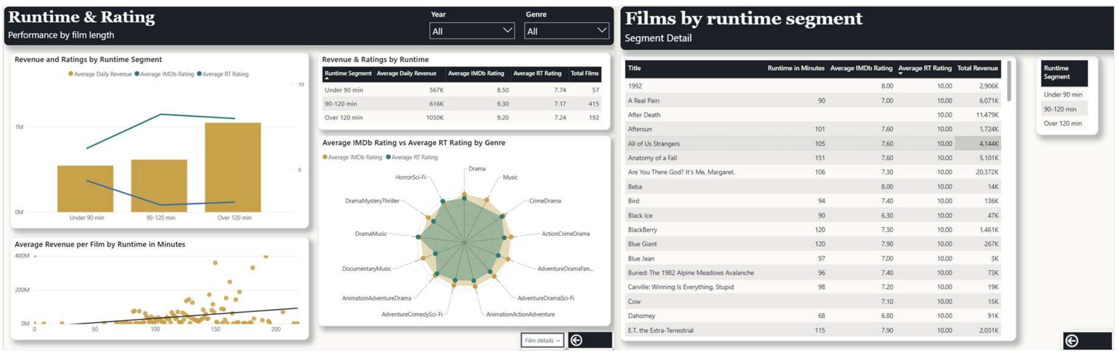

Figure 39: Report A (Performance & Return) - Runtime & Ratings (D3) page.

Page 4: Budget & Return (D8 + D5). D8 is answered with a ROI matrix crossing the Top-10 genres with Budget Tier, formatted as a heat-map (the Unknown tier is excluded), alongside an average-ROI-by-tier bar. A heat-map is chosen because ROI across a genre × budget-tier grid is a two-dimensional comparison, and conditional background color turns a dense matrix of numbers into an immediate read of where capital efficiency concentrates, far faster than scanning a table. The page carries KPI 2 - ROI as a conditional-format card to two decimals, with the color threshold set to the observed overall ROI of 0.92 (green at or above, red below), so each value is judged against the portfolio's real benchmark rather than an arbitrary one. D5 is answered on the same page, where it belongs, with a Top-5 table (Title, Total Revenue, and IMDb and Rotten Tomatoes ratings) ranked by IMDb and driven by a Budget Tier slicer, so the highest-rated films and their revenue can be compared within each tier.

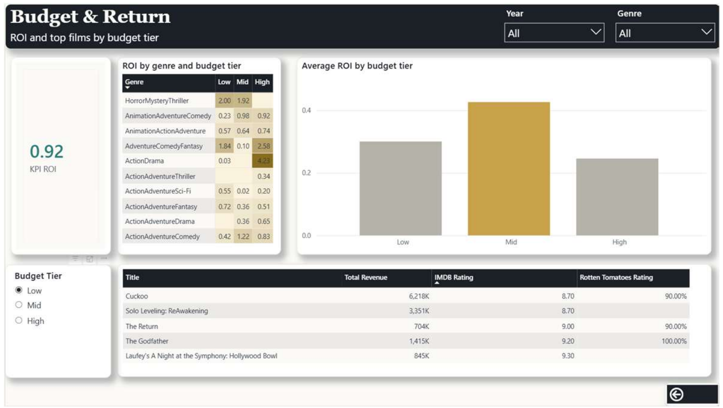

Figure 40: Report A (Performance & Return) - Budget & Return (D8, D5) page.

Page 5: Talent Value (D9). D9 is answered with a master bar of average revenue per film by lead actor (Top-15, restricted to actors with at least two films and to non-blank surnames), a "volume versus value" combo that places Total Films as columns against average revenue per film as a line, and a detail table; the genre dimension of D9 is obtained through the report's Genre slicer. A ranked bar is the natural visual for a ranking question, and the volume-versus-value combo is deliberate: it separates how many films an actor has from how much each earns, so a prolific actor is not mistaken for a high-value one. The page carries KPI 3 - Revenue vs Actor Average, read by selecting an actor, which likewise shows blank until an actor is chosen. The two-film threshold and the non-blank filter are deliberate responses to data limitations rather than cosmetic choices: few actors appear in multiple films, and some records carry no actor name, and leaving them in would distort the ranking.

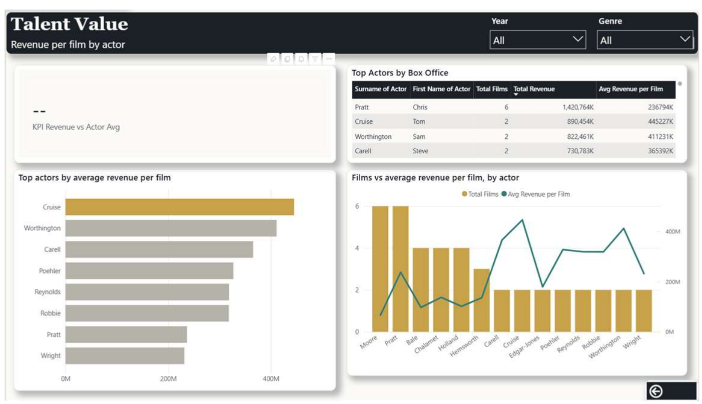

Figure 41: Report A (Performance & Return) - Talent Value (D9) page.

Every content page (2–5) carries a "← Overview" button back to the hub, so the committee can always return to the portfolio view in one click.

Business Insight. Read together, Report A tells the committee that return on a film is concentrated, not uniform: the average title returns only about $0.92 of box office for every $1 of budget, so where money is placed matters far more than how much is spent. The ROI matrix locates that return precisely: the mid-budget tier is the most capital-efficient overall, but the sharpest efficiency sits in a few specific low-budget genre blends (for example ComedyFantasy at roughly 9.9× and HorrorSci-FiThriller at 7.9×), not in broad genres such as plain Comedy. Revenue also scales with length: films over 120 minutes earn close to double the daily revenue of shorter ones, while the 90–120-minute band is both the most common and the best-rated, marking it as the safe commercial middle. The page-2 split between total and per-day revenue shows further that the genres earning the most in aggregate are not always those earning the most per screening day, and the talent view confirms that per-film value is carried by a small set of actors: volume is not value. In short, Report A turns the question "which films make money" into "which genre, budget and length combinations make money efficiently", which is the foundation for every allocation decision that follows.

6.3 Standard Report B — Timing & Geography

Report B answers the questions about when and where revenue is generated (D4, D6, D7) and hosts KPI 4. It is organized as a four-page hub-and-spoke report, following the same design intent as Report A: the visual on each page is chosen to make its question fastest to read, and the page is kept clean enough that the answer is obvious.

Page 1: Overview. The hub presents Total Revenue, ROI, and Total Films cards (ROI shown neutral, without color) above a master line of total revenue by month, with a small filled-map teaser and three navigation buttons to the Weekly, Timing, and Geography pages. The master visual is a line because the hub's job is to show the shape of revenue over time before any drill-down, and a line reads a trend at a glance; the map teaser previews the geographic dimension, and the ROI card is left neutral here because, as in Report A, there is no benchmark to beat at portfolio level. (screenshot: Report B Overview)

Figure 42: Report B (Timing & Geography) – Overview page

Page 2: Weekly Momentum (D4). D4 is answered with a line of total revenue by month, legended by the Top-5 genres, beside a bar of the Top-10 genres by average weekly revenue change. A line is used for the monthly series because revenue over time is a trend, and it is limited to the Top-5 genres so the chart stays readable rather than overplotting every genre; the momentum bar is a ranked comparison, so a bar with a conditional color rule, threshold set to the average weekly change of approximately $1.82M (green at or above, red below), makes the leading and lagging genres pop instantly. We document the caveat openly: in the observed build-up window all weekly changes are positive, so the measure is read as relative momentum between genres rather than as absolute decline and stating this stops a reader misreading a red bar as a loss.

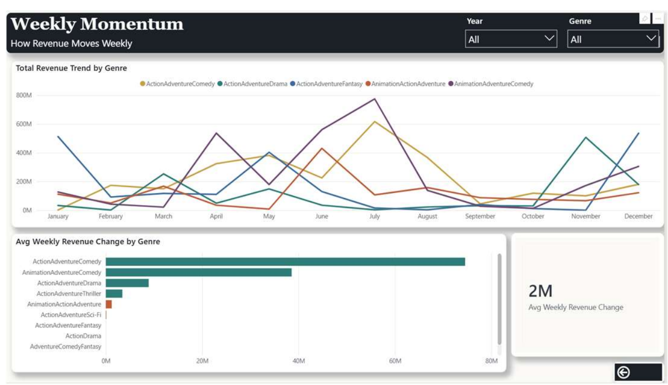

Figure 43: Report B (Timing & Geography) - Weekly Momentum (D4) page

Page 3: Release Timing & Audience Behavior (D6). D6 is answered with columns of average daily revenue by weekday, which surface the weekend (Friday–Sunday) concentration, and a decay curve of average daily revenue against Days Since Release, filtered to ≤ 70 days to remove the re-release outliers of older films that otherwise appear at around 20,000 days.

Columns are the right visual for the weekday comparison because they make the weekend spike immediately visible, and the decay curve is a line because revenue against time-since-release is a trend. The page carries KPI 4 - Days vs Genre Average, read per genre (zero means the genre sits at the average run length), which shows blank until a genre is selected in the slicer.

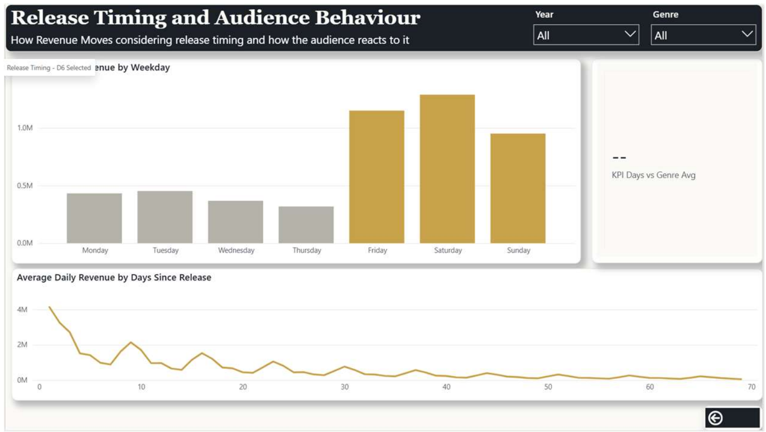

Figure 44: Report B (Timing & Geography) - Release Timing and Audience Behavior (D6) page.

Page 4: Geography of Production (D7). D7 is answered with a filled map of Main Filming Country (data category set to Country/Region, color saturation driven by total revenue), a bar of total revenue by the continent of the production company, and a detail table of country, total revenue, and total films. A filled map is the natural choice because geography is a spatial question, and color saturation encodes revenue magnitude, so the highest-earning countries stand out at a glance; the continent bar complements it with an aggregated ranking the map alone does not give, and the detail table carries the exact figures. A custom tooltip page (D7 Tooltip) showing the Top-5 titles by revenue is wired to the map and appears on hover, keeping the map uncluttered while still offering film-level detail on demand

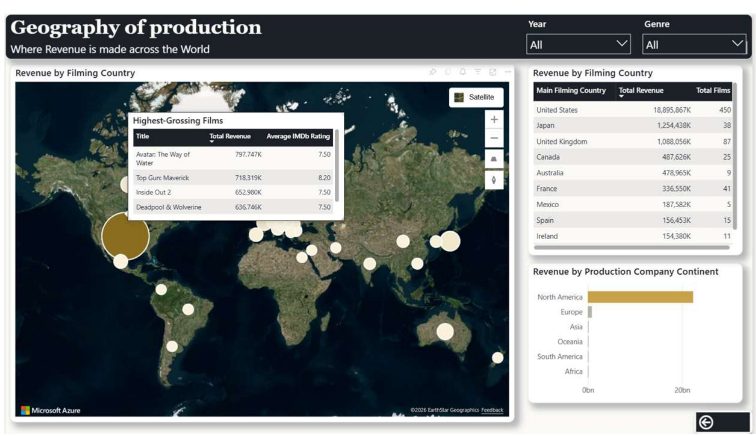

Figure 45: Report B (Timing & Geography) - Geography of production (D7) page.

Each content page carries a "← Overview" button back to the hub.

Business insight. Report B tells the committee that revenue is concentrated in time and in place, not spread evenly. In time, two patterns stand out: takings cluster heavily on the weekend (Friday–Sunday), and a film's commercial life is short: average daily revenue decays sharply after release and is largely spent within roughly ninety days, after which a title contributes little. This makes the release date and the opening window levers in their own right, since the same film earns more or less depending on when it opens and how its first couple of months are managed. The weekly-momentum view adds that genres differ in how their revenue builds, so momentum is itself a selection signal, read as relative strength between genres rather than as decline. In place, revenue concentrates in a limited set of filming countries and production-company regions rather than being globally uniform, giving MAD a geographic lens on where commercially successful production tends to originate. The takeaway is that timing and geography are returns levers MAD controls largely independently of a film's content — decisions about when and where to release and produce that can lift the return the content alone would not change.

6.4 Standard report C — Box-Office Forecasts (predictive analytics)

This report is the predictive component of the platform and deliberately goes beyond descriptive analysis. It gives visual form to the two machine-learning models documented in the Predictive Methodology: a per-genre linear regression of revenue trends (table ml_genre_trends) and a tree-ensemble (Gradient Boosting) classifier of box-office tier (table ml_tier_predictions), both materialized as Delta tables in the Lakehouse and consumed in Direct Lake by the semantic model. It answers business questions P1 (which genres are projected to grow or decline, and with what revenue) and P2 (what box-office tier is predicted from a film's profile), and on a fourth page it synthesizes both into an investment-priority view.

The report rests on a conceptual frame worth stating explicitly, because it answers the first question one would ask: why "predict" the tier of films that have already been released and whose box office is already known? Because the 782 historical films are not the object of the prediction: they are the training and validation base. The models learn the pattern that links a film's profile (budget, ratings, genre, age classification, release season, awards) to its box-office outcome, and we measure how well they reproduce known outcomes. The business value lies in applying that validated pattern to future or in-planning films, before MAD commits capital. Because a dashboard cannot run the model in real time on a film that does not yet exist, Report C does two things: it exposes the validated pattern (what distinguishes a High film) and it proves that the pattern holds; and it synthesizes the decision at genre level, because genres persist into the future (a specific past film is not investable, but "making a film of the AnimationAdventureComedy type" is a real, recurring decision). This frame is what makes a prediction over historical films defensible.

Integration into the semantic model. The ml_tier_predictions table joins to D Film through a 1:1 relationship on bk_film, a dimension extension that lets the prediction attributes (and the film poster) be used alongside the film dimension in any visual, without altering the governed warehouse dimension. The ml_genre_trends table joins to D Film by Genre (1:*), propagating genre selection across pages. These relationships are what make the slicers, tooltips and cross-filtering described below work.

Page 1: Overview. The hub of the report. Four header cards give the immediate read: Total Films (782), High-Tier Films (312), Genres Forecast to Grow (28, with Trend = Growing and n_years ≥ 3) and Average Model Confidence (83%). The master visual is a Key Influencers (AI) visual that answers the investment question directly ("what makes a film be predicted High?") surfacing budget as the dominant factor, followed by the Rotten Tomatoes and IMDB ratings, and exposing the Top Segments of profile where the probability of High concentrates. In a single screen the page gives the scale of the portfolio, the size of the opportunity, and the levers where the investable signal sits.

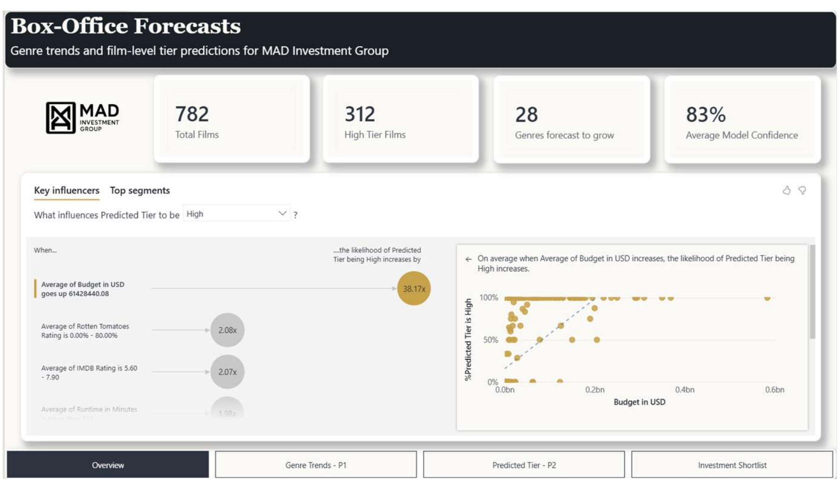

Figure 46: Report C (Box-Office Forecasts) – Overview page.

Page 2: Genre Trends & Projection (P1). This page answers P1. The central visual is a diverging dot/lollipop chart (a certified custom visual) of Annual Growth % per genre (green for genres that are growing, red for those declining) sorted descending. A diverging dot chart is the right visual here because projected growth is a deviation around zero: with green dots above the zero line and red below, the committee reads both the direction and the magnitude of each genre's trend at a glance, which a plain bar chart would flatten. A "Genres by trend" donut summarizes the composition, a line shows historical revenue for the leading genres over time, and a Trend slicer focuses the read. On this reliable (n_years ≥ 3) basis, the steepest projected growth is in ActionComedyDrama and the steepest decline in DramaMysteryRomance, the two ends of the chart. A material data limitation shaped this page, and we record it openly. The source covers only three years (2022–2024), and a genre's trend is an average direction over time: a single year is one data point, from which growth or decline simply cannot be estimated — at least two years are needed to fit a slope, and three to make it reliable. In practice only 56 of the 193 genres have all three years, while 95 have a single year (and are therefore forced to "Stable") and 42 have two. To avoid reading these forced-Stable genres as a real signal, we filter the P1 visuals to genres with the full three years (n_years ≥ 3); unfiltered, the "Stable" group dominates as an artefact of the data, not a market reality.

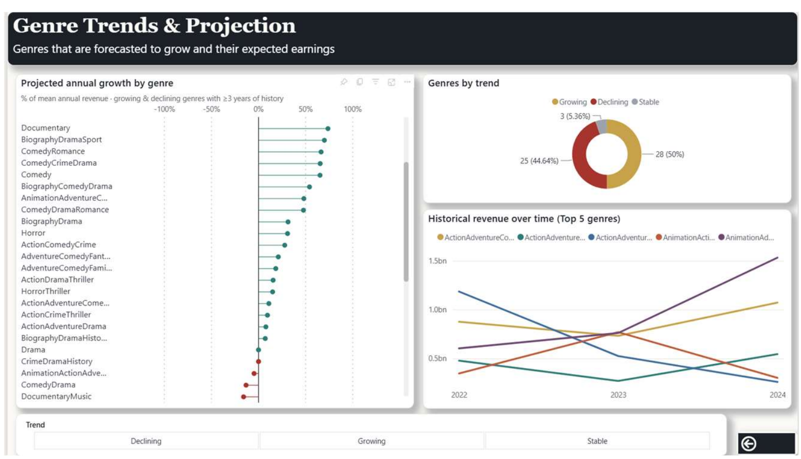

Figure 47: Report C (Box-Office Forecasts) - Genre Trends & Projections (P1) page.

Page 3: Predicted Box-Office Tier (P2). This page answers P2. A "Films by predicted tier" donut shows the composition (312 High / 249 Mid / 221 Low). Beside it, a column chart of average revenue per film by predicted tier serves as a validation visual: it confirms that the model's tiers track real revenue (High ≫ Mid ≫ Low), proving the predicted classes are economically meaningful and not arbitrary A detail table lists each film with Genre, Runtime

Segment, Release Season, Age Classification, Predicted Tier, Tier Confidence and Total Revenue, and hovering a film shows its poster in a tooltip, while a card reports the average model confidence (≈ 83%). Four profile slicers (Genre, Runtime Segment, Release Season, Age Classification) are what turn the page from a summary into a decision tool: the committee builds a hypothetical film's profile and reads the tier and confidence the model would assign — which is exactly how the model is meant to be used on a film that does not yet exist. The tier model is trained on ten features (genre, age classification, release season, runtime, budget, IMDB, Rotten Tomatoes, awards won, awards nominations and film release year). One methodological point is recorded here: the predictions and confidence for the historical set are computed out-of-fold (cross_val_predict, stratified cross-validation), not in-sample. Scoring each film with a model that has already "seen" it inflates apparent accuracy toward 98%; out-of-fold scoring (where each film is classified by folds that excluded it) returns the honest figure. The resulting average confidence (≈ 83%) sits slightly above the validated test accuracy, which is normal for a well-regularized tree ensemble. We deliberately distinguish the two: accuracy is the model's validated hit rate; confidence is the certainty the model assigns to each prediction, and the dashboard card reports the latter.

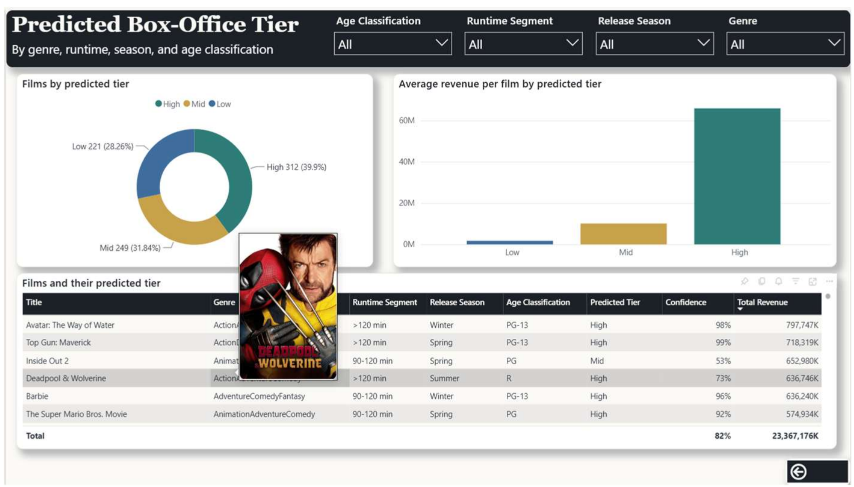

Figure 48: Report C (Box-Office Forecasts) - Predicted Box-Office Tier (P2) page.

Page 4: Where to Invest. This is the page that converts the two models into an investment decision. It answers: given which genres are growing (P1) and which genres produce High films (P2), where should MAD allocate capital? The central visual is a genre opportunity map (a scatter) with one bubble per genre: Annual Growth % on the X axis (momentum), Hit Rate (the fraction of the genre's films predicted High, i.e. the probability of success) on the Y axis, Projected Revenue as the bubble size (the size of the opportunity) and Trend as the bubble colour (green growing, red declining). Two quadrant lines (X = 0, separating decline from growth, and Y ≈ 0.40, the overall base rate of High films) divide the space; the top-right quadrant, with large green bubbles, is the sweet spot: growing genres, with an above-base-rate success rate and a large opportunity. It is, in effect, a four-dimensional risk-reward map of investment by genre. Beside it, a "Top Target Genres" table gives the ranked shortlist (Genre, Annual Growth %, Hit Rate, Projected Revenue and High Films) filtered to n_years ≥ 3 and sorted by projected revenue, with the High Films column doubling as a sample-size check (a 100% hit rate on five films is treated as indicative, not firm). A detail panel shows example films of the selected genre (it responds to a click on a bubble or row) with the poster tooltip. The synthesis is at genre level by design, because a specific historical film is not investable, but the genre persists and is the unit at which MAD chooses the type of film to back. Read from the data, clear targets emerge, growing genres with high, well-sampled success rates: AnimationAdventureComedy (≈ 1.9 billion projected revenue, 63% hit rate, +48% growth, 17 films), ActionAdventureComedy (≈ 1.09 billion, 74%, +11%, 14 films) and ActionAdventureDrama (90% hit rate, 9 films). In a single decision view the page crosses momentum, probability of success and size of opportunity.

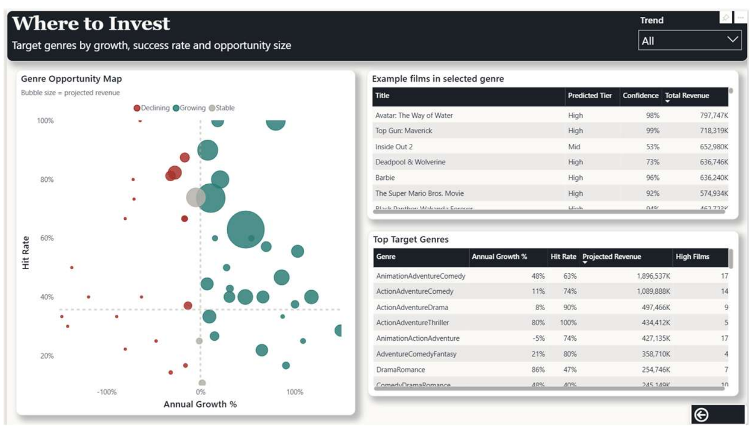

Figure 49: Report C (Box-Office Forecasts) - Where to Invest page

Business insight. Report C turns the platform from a record of what happened into a tool for deciding what to fund next. Two findings carry the most weight. First, the investable signal is concentrated: of everything that distinguishes a High-tier film, budget is by far the dominant driver, followed by critical quality (Rotten Tomatoes and IMDb), so a film's funding level and expected reviews are the levers that move its predicted outcome most; and because the predicted tiers track real revenue (High > Mid > Low), MAD can trust those predictions rather than treat them as a black box. Second, opportunity is uneven across genres: only a minority are projected to grow, and the genre opportunity map isolates the few that combine momentum, a high success rate and a large projected revenue (AnimationAdventureComedy, ActionAdventureComedy and ActionAdventureDrama) as the priority shortlist, while flagging thin-sample cases so they are read as indicative rather than firm. The practical value is the shift from hindsight to foresight: rather than judging a film only after release, MAD can screen a planned film's profile against the model, read its likely tier and confidence before committing capital, and concentrate that capital on the growing, high-hit-rate genres the data rewards.

6.5 Paginated reports: Talent league tables (answering D2)

We built two Fabric paginated reports in Power BI Report Builder, "Lead Actors by Cumulative Box-Office Revenue" and "Lead Directors by Cumulative Box-Office Revenue", each a Top 30 ranked by cumulative box-office revenue. We moved to Report Builder because the Fabric web editor could not sort, remove the empty-name row, or cut to a Top N; the reports still connect live to the DW_MAD_MOVIES semantic model and publish back to the same Fabric workspace, so they remain Fabric paginated reports.

Why this design. A ranking of thirty names is best shown as a fixed-layout table rather than a chart, and the paginated format makes it print-ready and exportable: a league table the committee can circulate, which is what paginated reports are for and the interactive reports are not.

Table columns. Each report renders the same table: a rank (#), the name (Actor or Director), Nationality, Films (the Total Films measure), Avg Revenue / Film and Total Revenue (USD). The rows are the Top 30 sorted by Total Revenue (USD) descending, with the empty-name row excluded. A single DAX dataset produces the ranking, the sort and the Top 30 cut, reusing the measures and currency formatting already defined in the model.

The directors dataset is identical over the D Director dimension. The report concatenates First Name and Surname into the single name column shown, and Report Builder adds the rank (#) as a row number.

How they answer D2 and reinforce D9. D2 asks which lead actors and which lead directors rank highest by cumulative box-office revenue across all their films. The two reports answer it directly, one ranked list each, ordered by cumulative Total Revenue. Cumulative revenue is the right lens for D2 because it captures the total commercial weight a name has carried; it complements D9 (Talent Value, in Report A), which ranks talent by average revenue per film.

Lead Direcctors by Cumulative Box-Office Revenue - Top 30

MAD Investment Group · Business Question D2

#	Director	Nationality	Films	Avg Revenue / Film	Total Revenue (USD)
1	James Cameron	Canada	4	$209,503,789	$838,015,155
2	Joseph Kosinski	USA	1	$718,318,561	$718,318,561
3	Christopher Nolan	England	4	$179,489,958	$717,959,830
4	Kelsey Mann	Unknown	2	$326,938,692	$653,877,384
5	Shawn Levy	Canada	1	$636,745,858	$636,745,858
6	Greta Gerwig	USA	2	$318,188,020	$636,376,040
7	Michael Jelenic	USA	2	$293,281,248	$586,562,495
8	Ryan Coogler	USA	1	$462,723,023	$462,723,023
9	Sam Raimi	USA	1	$411,331,607	$411,331,607
10	Justin K. Thompson	USA	2	$190,796,877	$381,593,754
11	Colin Trevorrow	USA	1	$376,009,080	$376,009,080
12	Kyle Balda	USA	1	$369,779,140	$369,779,140
13	Chris Renaud	USA	1	$361,004,205	$361,004,205
14	James Gunn	USA	2	$179,526,333	$359,052,666
15	Taika Waititi	New Zealand	2	$174,985,328	$349,970,656
16	Jeff Fowler	USA	2	$171,215,899	$342,431,798
17	David G. Derrick	Unknown	1	$318,363,048	$318,363,048
18	Ron Clements	USA	1	$298,172,056	$298,172,056
19	Tim Burton	USA	1	$294,072,781	$294,072,781
20	Denis Villeneuve	Canada	2	$141,913,463	$283,826,926
21	Joel Crawford	Unknown	2	$140,049,059	$280,098,118
22	Lee Isaac Chung	USA	1	$267,762,265	$267,762,265
23	Jon Watts	USA	2	$128,268,998	$256,537,996
24	Jon M. Chu	USA	1	$250,921,435	$250,921,435
25	Paul King	Unknown	1	$235,496,712	$235,496,712
26	Peyton Reed	USA	1	$214,504,909	$214,504,909
27	Alejandro Monteverde	Mexico	2	$101,844,930	$203,689,859
28	James Mangold	USA	2	$101,491,154	$202,982,308
29	David Leitch	USA	3	$65,501,852	$196,505,556
30	Adam Wingard	USA	1	$196,350,016	$196,350,016

Source: DW_MAD_MOVIES semantic model · Page 1 of 1

Lead Actors by Cumulative Box-Office Revenue - Top 30

MAD Investment Group · Business Question D2

#	Actor	Nationality	Films	Avg Revenue / Film	Total Revenue (USD)
1	Chris Pratt	USA	6	$236,794,076	$1,420,764,458
2	Tom Cruise	USA	2	$445,226,972	$890,453,944
3	Sam Worthington	England	2	$411,230,628	$822,461,256
4	Steve Carell	USA	2	$365,391,673	$730,783,345
5	Amy Poehler	USA	2	$326,938,692	$653,877,384
6	Ryan Reynolds	Canada	2	$318,374,442	$636,748,883
7	Margot Robbie	Australia	2	$318,148,656	$636,297,311
8	Timothée Chalamet	USA	4	$136,956,370	$547,825,478
9	Letitia Wright	Guyana	2	$231,464,079	$462,928,158
10	Benedict Cumberbatch	England	1	$411,331,607	$411,331,607
11	Tom Holland	England	3	$135,062,272	$405,186,816
12	Chris Hemsworth	Australia	2	$201,155,500	$402,311,000
13	Christian Bale	England	4	$97,003,985	$388,015,941
14	Shameik Moore	USA	2	$190,796,877	$381,593,754
15	Daisy Edgar-Jones	England	2	$178,996,513	$357,993,025
16	Cillian Murphy	Ireland	2	$165,661,881	$331,323,761
17	Auli'i Cravalho	USA	1	$318,363,048	$318,363,048
18	Jodi Benson	USA	1	$298,172,056	$298,172,056
19	Alec Baldwin	USA	1	$294,072,781	$294,072,781
20	Antonio Banderas	Spain	3	$93,557,220	$280,671,661
21	Lupita Nyong'o	Mexico	2	$139,511,814	$279,023,628
22	Cynthia Erivo	England	1	$250,921,435	$250,921,435
23	Paul Rudd	USA	1	$214,504,909	$214,504,909
24	Keanu Reeves	Lebanon	3	$65,963,470	$197,890,411
25	Rebecca Hall	England	1	$196,350,016	$196,350,016
26	Will Smith	USA	2	$96,992,720	$193,985,439
27	Jack Black	USA	2	$96,941,788	$193,883,575
28	Ben Schwartz	USA	1	$190,872,904	$190,872,904
29	Neve Campbell	Canada	2	$94,901,397	$189,802,794
30	Mckenna Grace	USA	4	$46,534,094	$186,136,376

Source: DW_MAD_MOVIES semantic model · Page 1 of 1

Figure 50: Talent league tables paginated reports, ranking the Top 30 actors and Top 30 directors by cumulative box-office revenue (D2).

Business insight. The two tables make the cumulative-versus-per-film distinction concrete. The actor ranking is led by Chris Pratt on cumulative revenue, but that total is built on six films, whereas Tom Cruise is second on cumulative revenue with the highest average per film, from only two. A name can therefore top the total ranking through volume while another delivers far more per individual film, which is exactly why the report shows Avg Revenue / Film and Total Revenue side by side. For a casting or greenlight decision the committee reads the two together: cumulative revenue points to the bankable names with a long commercial track record, while the per-film average flags the names that most lift a single film's expected return.

7. Executive dashboard, "MAD Movies"

The three reports are exploration surfaces, built hub-and-spoke for an analyst drilling into a theme; on top of them we built a single Power BI dashboard in Fabric, "MAD Movies", a one-canvas surface that pins the decision-critical tiles from the three reports into one place. It is the committee's front page, built for a decision rather than for exploration, and it is a genuine Power BI dashboard artefact (pinned tiles from several reports) and therefore distinct from, and complementary to, the reports' own Overview pages. Because a dashboard has no pages or navigation, the story is carried by layout and visual hierarchy: the canvas is read top to bottom and left to right, organized into five labelled sections that move from context to recommendation, with the Genre Opportunity Map placed in the focal position because it is the answer to the business question. We pinned only glance-able, static tiles: a dashboard tile is a frozen snapshot, so interactive or drill-dependent visuals (such as the Key Influencers visual) were deliberately left in the reports, where they keep their purpose.

The pinned tiles, and the reports they come from, are:

- **The Portfolio at a Glance**: the context KPI cards (Total Revenue, ROI 0.92, Total Films and Average Revenue per Film).
- **What Drives Returns**: Average ROI by budget tier and the Genre × Budget Tier ROI matrix, both from Report A (Performance & Return).
- **When & Where Revenue Is Made**: Revenue by Weekday, the Days-Since-Release decay curve and Revenue by Filming Country, all from Report B (Timing & Geography).
- **The Recommendation**: the Genre Opportunity Map and the Top Target Genres shortlist from Report C (Box-Office Forecasts), plus Top actors by average revenue per film from Report A.
- **Going Deeper**: Revenue by Month (Report B), Revenue & Ratings by Runtime (Report A) and the Genres-by-trend composition (Report C).

The dashboard exposes only context KPIs (Total Revenue, ROI, Total Films, Average Revenue per Film), which are valid without any selection; the four benchmark "vs. average" KPIs stay in the reports, where a selection makes them meaningful. Every tile links back to its source: clicking a pinned tile opens the report page it was pinned from, so the committee can move from the at-a-glance canvas into the interactive report for the detail behind any number. The canvas opens with framing copy: the title "Where to Invest Next" and a subtitle that frames the 782 films of box-office history as MAD's next decision.

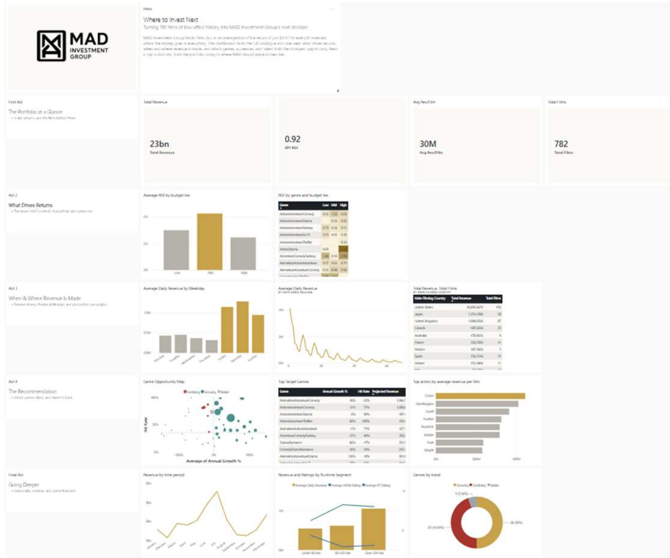

Figure 51: The "MAD Movies" executive dashboard ("Where to Invest Next"), consolidating the decision-critical tiles from the three reports onto a single canvas.

8. From Analysis to Decision: The Ideal Film Profile

The eleven Business Questions, answered separately across the reports, converge on one practical question for MAD Investment Group: what does the most fundable film look like? This section synthesises the findings into a recommended profile and three concrete plays, one per budget tier, so the analysis ends in a decision rather than a description. Where the "Where to Invest" view works at genre level, this blueprint combines every dimension: genre, budget, audience, runtime, timing, quality and talent.

What the analysis says. Five findings dominate:

1. **Return is concentrated, not uniform.** The average film returns about $0.92 of box office per $1 of budget (ROI ≈ 0.92), so the edge comes from selection, not spending.
2. **At low budget, precision beats breadth.** Specific blends deliver outstanding ROI (ComedyFantasy 9.89, HorrorSci-FiThriller 7.92, BiographyCrimeDrama 6.37) while plain Comedy (0.28) or Horror (1.08) do not; the gain is a precise genre, not a broad one.
3. **Longer films earn more.** Average daily revenue rises from $567K (under 90 min) to $616K (90 to 120 min) to $1.05M (over 120 min); the 90 to 120 minute band is also the most common, at 415 films.

4. **Once budget is set, ratings are the next lever.** Of the modelled drivers of a top-tier film, budget is the strongest, followed by critical quality (Rotten Tomatoes and IMDb).

5. **Timing and momentum are consistent.** The growing, well-sampled genres are AnimationAdventureComedy, ActionAdventureComedy and ActionAdventureDrama, and revenue concentrates on weekend releases within a roughly 70-day theatrical window.

Three disciplines apply at every budget: high IMDb and Rotten Tomatoes ratings, a weekend release, and a roughly 70-day theatrical window. The rest of the profile changes with the budget:

	Low budget	**Mid budget**	**High budget**
Goal	Maximize ROI on small capital	Best balance of ROI and revenue	Maximize absolute revenue
Genre play	High-ROI blends: ComedyFantasy, HorrorSci-FiThriller, BiographyCrimeDrama	Growing target genres: AnimationAdventureComedy, ActionAdventureComedy	Large-scale Action/Adventure (proven High-tier cells)
Runtime	≤120 min, to keep cost down	90 to 120 min, the most common band	Over 120 min, highest daily revenue (~$1.05M)
Audience	Genre fan-base	Broad / family	Broad / four-quadrant
Talent	High per-film-value actors that fit the budget (D9)	Proven-draw actors (D2 / D9)	Top cumulative-revenue draws (D2)
Expected return	Highest ROI, smallest revenue	Strongest ROI-to-revenue balance	Lower ROI, highest revenue

Table 5: Recommended film profile by budget tier (low, mid, high).

The recommendation is a portfolio of three distinct plays, not one template: tightly-defined high-ROI genres at low budget (where six-to-ten-times-budget returns are achievable), a broad, well-rated film in a growing genre as the efficient mid-tier core, and long-form Action/Adventure tentpoles at high budget that trade ROI for absolute revenue. What unifies them is the first finding: because the average film barely breaks even, value comes from matching the profile to the budget the data rewards, not from spending more.

9. Extra work

This section gathers, in one place, the developments that go beyond the project's standard reporting techniques, so that they are unambiguous to identify and credit. Each item states what it is and where it can be seen in the reports.

9.1 A predictive layer integrated into the governed semantic model

Report C adds a supervised machine-learning layer and, more unusually, feeds it back into the reporting model itself. A Gradient Boosting classifier predicts a film's box-office tier (Low, Mid or High) from ten features, and a companion per-genre linear regression projects revenue trends on a reliable basis (genres with at least three years of data). Two choices make this more than a one-off model run. First, the on-screen confidence is validated out-of-fold using cross_val_predict with stratified cross-validation, so each film is scored only by folds that never saw it and the reported confidence (≈ 83%) is honest rather than the inflated value a single in-sample fit would produce. Second, the predictions are not left as a standalone file but materialized as Delta tables and joined into the governed model: ml_tier_predictions connects to D Film through a 1:1 dimension-extension relationship on bk_film, so the predicted tier, the confidence and each film's poster flow into report visuals alongside the warehouse dimension without altering it. The visible results are the P1 and P2 pages and the average-revenue validation chart.

9.2 Bespoke "Genre Opportunity Map" & prescriptive "Where to Invest" view

The original contribution on the Where to Invest page is what we encode on the scatter and the decision it produces. The genre opportunity map carries four dimensions at once: Annual Growth % on the X axis for momentum, Hit Rate (the share of a genre's films predicted High) on the Y axis for probability of success, Projected Revenue as the bubble size for size of opportunity, and Trend as the bubble color. Two base-rate reference lines, one vertical at zero growth and one horizontal at the overall rate of High films (≈ 0.40), turn the plane into a risk-reward quadrant map. Combined with a ranked Top Target Genres shortlist, it converts the two models' outputs into a single, prescriptive capital-allocation view rather than a descriptive chart.

9.3 A custom marketplace visual

Two visuals come from the marketplace rather than the built-in set. On the P1 page of Report C, a diverging dot/lollipop chart (a certified custom visual) presents Annual Growth % per genre, colored green or red by trend and sorted descending; it was chosen because projected growth is a deviation around zero, which a deviation-style dot chart reads far more clearly than a bar. On the Runtime & Ratings page of Report A, a radar chart compares Average IMDb Rating against Average Rotten Tomatoes Rating across genres, using a shared radial scale to place both rating dimensions for many genres in a single multivariate view.

10. Critical Review / Lessons Learned

Building this platform taught the group as much about the realities of business intelligence as about Power BI, and the lessons that matter most came from the difficulties we worked through.

The platform shapes the solution as much as the visuals do. Direct Lake and the Fabric web editor (no calculated columns, no Edit interactions, no native drill-through) pushed our binning upstream into SQL and made us engineer our own interactions, and those constraints are where our most original work came from.

Honest analysis surfaces data artefacts rather than hiding them. The apparent "stable majority" of genres was an artefact of only three years of data and vanished once we filtered to reliable ones; the data's shape can manufacture false conclusions, and the honest response is to show the defensible view, not the misleading one.

A model must be validated honestly and explained before it is trusted. Out-of-fold scoring replaced a flattering 98% accuracy with the real figure and taught us to separate accuracy from per-prediction confidence; we also had to defend why predicting already-released films is legitimate at all, so the analytical narrative mattered as much as the numbers.

Delivery must fit its purpose. Paginated tables, interactive reports and a decision dashboard each suit a different use, and small DAX context errors can quietly distort results; a report is only as trustworthy as the model, the data discipline and the validation beneath its visuals.

11. Conclusions

From MAD Investment Group's perspective, the project delivers a single, connected BI platform that turns years of box-office history into investment-ready evidence. The value now available to the firm has four parts.

One governed source for every question. MAD now operates from a single Direct Lake semantic model over the DW_MAD_MOVIES star schema that answers all eleven Business Questions, so content, talent, timing, geography and budget are queried from the same source instead of reconciled across spreadsheets.

Direct answers to recurring decisions. The Performance & Return and Timing & Geography reports answer the questions the committee asks repeatedly (what content earns most, which talent carries the most commercial weight, how timing and geography shape revenue, and which genre-and-budget combinations are most capital-efficient), each tied to a specific Business Question rather than a one-off analysis.

From hindsight to foresight. The predictive report moves part of the investment process from looking back to looking forward: MAD can screen a planned film's likely box-office tier before committing capital (P2) and spot the genres gaining momentum ahead of the next release window (P1). The "Where to Invest" map crosses momentum, success probability and opportunity size into a ranked, ROI-aware shortlist, and the executive "Where to Invest Next" dashboard consolidates these views onto one canvas, with a click-through from any tile to the detail behind it.

Durable, reusable value. For a firm whose business is allocating capital under uncertainty, the value is decision support that is fast, consistent and grounded in its own data: the committee can move from a question to a defensible, ROI-aware answer in one branded environment, and re-run the models as new films are released without rebuilding anything. The platform does not remove the risk inherent in film investment, but it materially narrows the uncertainty around each decision, exactly the edge MAD set out to build.

Generative AI tools, including large language models, were used during this project to support methodology decisions, code development, and the drafting and revision of this report.